

Model Question Paper I**INTRODUCTION TO MICROSERVICES***Time: 3 Hour**Max. Marks: 75***PART A**

- I. Answer **all** questions in one word or one sentence. Each question carries **1 mark**.
(9 * 1 = 9 marks)

1	Define Microservices	M1.01	R
2	Define service discovery	M1.01	R
3	Define code coupling	M1.02	R
4	Define monolithic architecture	M2.01	R
5	Expand HTTP	M3.03	R
6	Define API	M3.01	R
7	List any two non functional requirements of microservices	M4.01	R
8	Define throughput	M4.02	U
9	Define JSON Web Token (JWT)	M4.03	R

PART B

- II. Answer any **eight** questions from the following, each question carries **3 marks**.
(8 * 3=24 marks)

1	Discuss the use of API Gateway	M1.01	R
2	Discuss the benefits of Microservices from business perspective	M2.04	U
3	Discuss the deployment issues in Monolithis architecture	M2.02	R
4	The cost for Monolithic infrastructure is high. Justify.	M2.02	U
5	Explaing interprocess communication methods in monolithic architecure	M3.01	R
6	Discuss the need of load balancing	M3.02	U
7	Discuss about design patterns which helps microservices service-to-service network calls more resilient	M2.02	A
8	Describe GET and POST	M3.03	U
9	Discuss about horizontal scalability and vertical scalability in microservices	M4.02	U
10	Define autheriation and authentication with example	M4.03	U

PART C

Answer **ALL** questions. Each question carries **7 marks**.

(6*7=42 marks)

III	Explain Microservice architecture with a diagram	M1.01	U
OR			
IV	Discuss the challenges of Microservices	M1.04	U
V	Explain the features of Microservices	M1.02	U
OR			
VI	Discuss the advantages of microservices	M1.03	U
VII	Explain the Monolithic architecture with an example	M2.01	U
OR			
VIII	Compare Monolithic and Microservices architecture	M2.03	U
IX	Discuss the technical benefits of Microservices	M2.04	R
OR			
X	Explain Synchronous communication in Monolithic architecture	M3.03	U
XI	Explain Asynchronous communication in Monolithic architecture	M3.03	U
OR			
XII	Explain the advantage of scaling in microservices	M4.02	A
XIII	Explain OAuth2 protocol	M4.03	A
OR			
XIV	Discuss the advantages of OAuth2	M4.03	U

